

ARIMA modelling

AR(p)
 $x_{t-1}, x_{t-2}, \dots, x_{t-p}$

Capturing the nature of the autocorrelation in the data

(fpp) Chapter 9:
Sections 9.1, 9.3–9.8

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Non-seasonal ARIMA models

- ARIMA models provide another approach to time series forecasting. Exponential smoothing and ARIMA models are the two most widely-used approaches in time series forecasting. They provide complementary approaches to the problem.
- While exponential smoothing models are based on a description of the trend and seasonality in the data, ARIMA models aim to describe the autocorrelation in the data.
- Thus, if we want to focus on fitting a model based simply on the pattern of the data (i.e. trend and seasonality) then we would look to Exponential smoothing models. However, if we are interested in finding a model that captures the nature of the autocorrelation in the data, ARIMA models are what we use.
- We have continuously emphasised that, in practice, many time series such as those related to socio-economic and business processes show non-stationary behaviour.
- ARIMA models are very useful for modelling a wide variety of stationary and non-stationary time series processes. ARIMA stands for 'AutoRegressive Integrated Moving Average' and refers to a family of models that are mixtures of autoregressive, AR(p) and moving average MA(q) models.
- Note that the moving average MA(q) models are NOT the same thing as moving average₂₁₄ smoothing that we considered previously

Non-seasonal ARIMA models

- If a non-stationary time series is modelled by both an AR(p) and a MA(q) process, we say that it is modelled by an ARIMA(p, d, q) process. It takes the form:

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$

AR(p) part
MA(q) part

- The "I" in ARIMA stands for "integrated", which basically means you're transforming the data by taking differences. (Integration is the reverse of differencing – which refers to the process of back-transforming your forecasts of your transformed data to the scale of the original time series.
- We say an ARIMA model has been "Integrated" at the order d (e.g., the dth difference) if we have had to apply a dth-order difference to the non-stationary time series in order to make it stationary.

The ARIMA model is typically written as ARIMA(p, d, q) where:

- p is the number of autoregressive terms
- d is the order of differencing
- q is the number of moving average terms

AR MA
 ↓ ↓
 ARIMA(1, 1, 3)
 ≠
 ARIMA(3, 1, 1)

- For example, an ARIMA(1,1,0) is a first-order AR model with one order of differencing. An ARIMA(2,1,1) refers to a mixed model consisting of two AR terms and one MA term that has been fitted to a time series that has been differenced once

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General Approach to Modelling a Time Series [that may have significant autocorrelation at some lag g]

The approach outline below serves as an outline to the general process to follow when performing a time series analysis when we suspect autocorrelation is present.

1. Plot the series and examine the main features of the graph, checking whether there is changing variance, a trend, a seasonal component, any apparent sharp changes in behaviour or any outlying observations.

TRANSFORM the series if it is non-stationary:

2. By stabilising the variance (if necessary) and
↳ Box-Cox transformation
3. Removing any trend /seasonal components via differencing at the appropriate lag to get a stationary series
↳ ndiffs(?)? nsdiffs(?)?

4. Choose an ARIMA(p,d,q) model to fit the transformed data/time series, based on the nature of the autocorrelation of the transformed data/time series, by making use of various sample statistics including plots of the ACF and PACF. Note that it is unlikely you will be able to specify THE best ARIMA(p,d,q) model from simply examining the ACF and PACF plots. In practice, it is common to:

- a. Estimate various ARIMA(p,d,q) models for a whole range of orders $0 \leq p \leq P$ and $0 \leq q \leq Q$
- b. Compare all the models by information criteria (AIC)

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General Approach to Modelling a Time Series [that may have significant autocorrelation at some lag g]

5. Estimate the parameters of that model (Computer software uses different methods available to do this e.g. MME, MLE) – **We do NOT concern ourselves with manually estimating the parameters in this course i.e. we let RStudio do the job!!!**

6. Perform diagnostic checks of the residuals of the model:

(i) Independence of error terms, (ii) Mean of 0, (iii) Normality of error terms, (iv) Constant error variance (Homoscedasticity)

7. Forecast future values of the time series using an appropriate fitted model.

NOTE: In practice, there may be a range of similar models (with different parameters) that may be appropriate for a given time series. Generally, we fit a few slightly different models and then compare the accuracy of the fit of each model **relative to the other models** using some form of information criterion (AIC) and measures of forecast accuracy (MSE/RMSE and MAE) to select the most appropriate model for forecasting purposes.

We will now proceed to consider each of the AR(p) and MA(q) models separately before discussing how to fit ARIMA(p,d,q) models

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Autoregressive, AR(p), models

- The observed value at time t depends linearly on the last p observed values, and the model looks like a regression model – hence the term autoregression
 - i.e. the future value of a variable is assumed to be a linear combination of p past observations and a random error together with a constant term.
- A process $\{X_t\}$ is said to be an autoregressive process of order p (abbreviated AR(p)) if:

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \phi_3 X_{t-3} + \dots + \phi_p X_{t-p} + \epsilon_t$$

$N(0, \sigma^2)$

- Where ϵ_t is a white noise process with mean 0 and variance σ^2 that represents the error series, and c is a constant related to the mean of the process. Note that often we deal with AR(p) processes that have a mean of 0, corresponding to $c = 0$.
- NOTE:** The β_t 's we used previously are replaced by the ϕ_t 's here in order to distinguish between the use of AR(p) models when a regression model has been fit, and the more general use we will be discussing here. ~~The parameters ϕ_t here are NOT the same as used in Holt's methods as a dampening parameter for the slope.~~

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Autoregressive, AR(p), models

Autoregressive series are important because they have a natural interpretation — the next value observed is a slight perturbation of a simple function of recent observations.

Autoregressive models are remarkably flexible at handling a wide range of different time series patterns, and have a wide application:

- The study of mortality rates
- GDP dynamics
- Climatological issues like the el nino effect and measuring solar radiation
- The study of volcanic tremors and brain electrical activity mapping
- They are frequently applied by financial investors

^{p=1}
The AR(1) process: The simplest example of an AR process is the first-order case, denoted AR(1), given by:

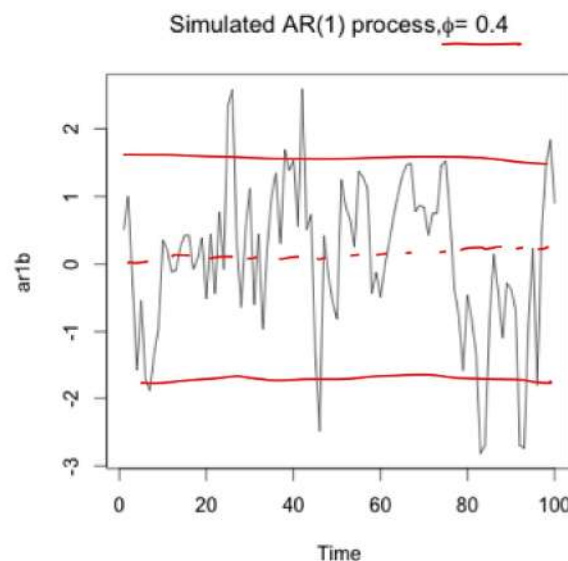
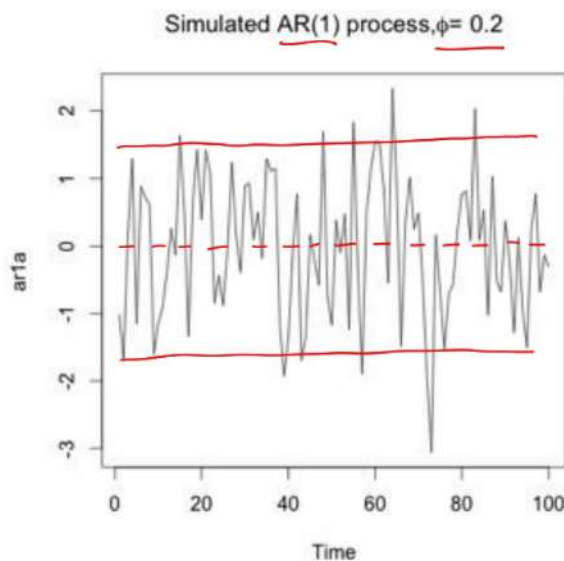
$$X_t = c + \phi_1 X_{t-1} + \epsilon_t$$

- We normally restrict autoregressive models to stationary data, in which case some constraints on the values of the parameters are required. The AR(1) process is stationary only when $|\phi_1| < 1$. As the value of ϕ_1 approaches either 1 or -1, the AR(1) process exhibits stronger non-stationary behaviour. $\phi_1 \rightarrow 1$: mean ?
 $\phi_1 \rightarrow -1$: variation ?
- Consider some examples of simulated AR(1) processes for different values of the coefficient ϕ_1 :

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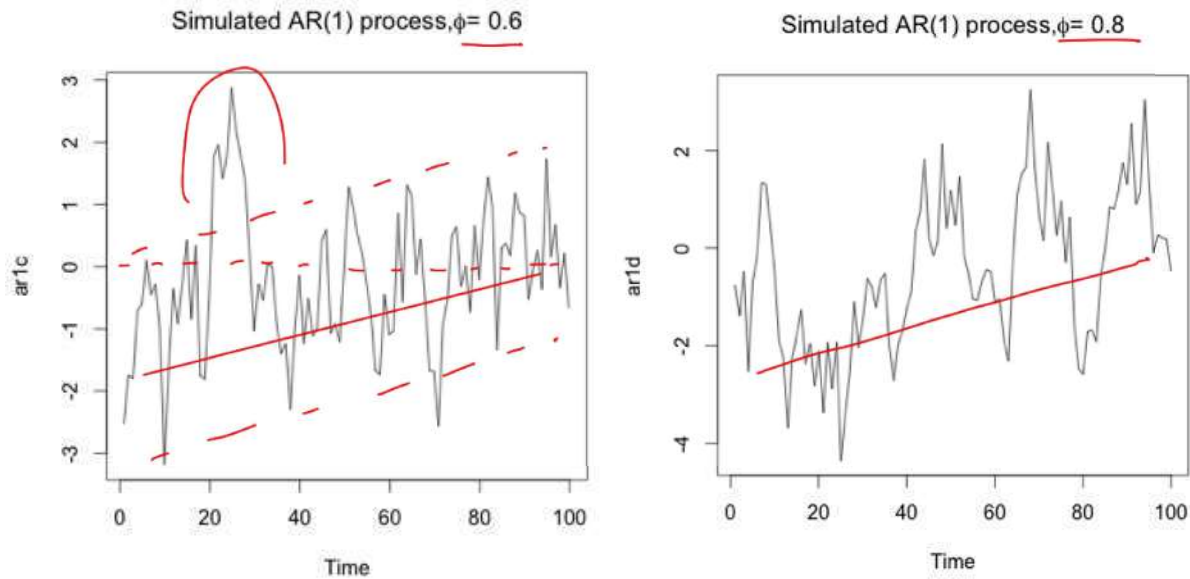
Autoregressive, AR(p), models

ϕ_1 positive



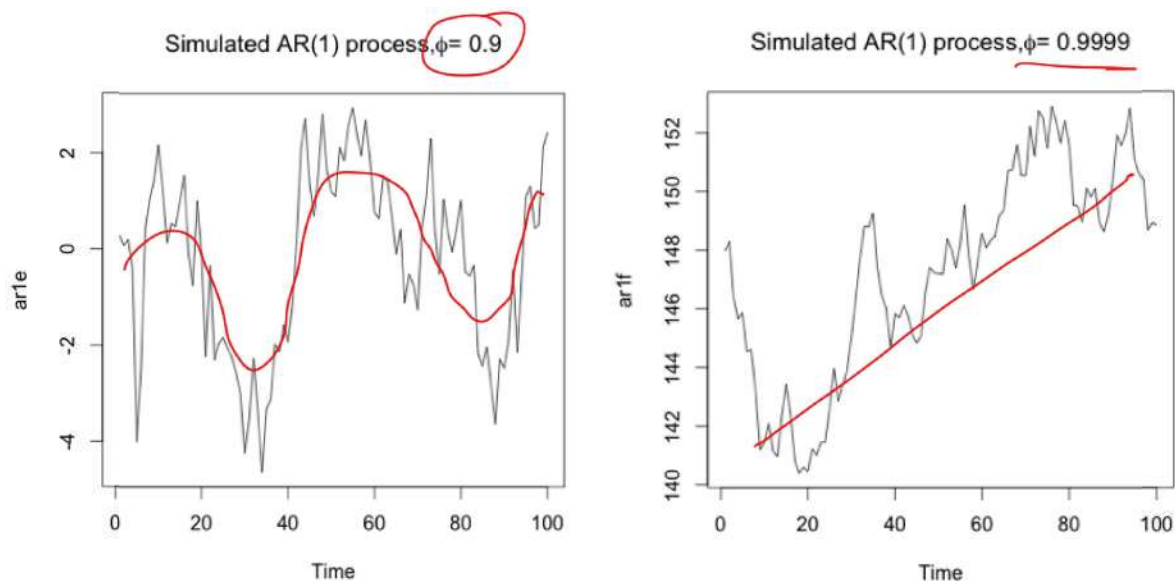
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Autoregressive, AR(p), models



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Autoregressive, AR(p), models

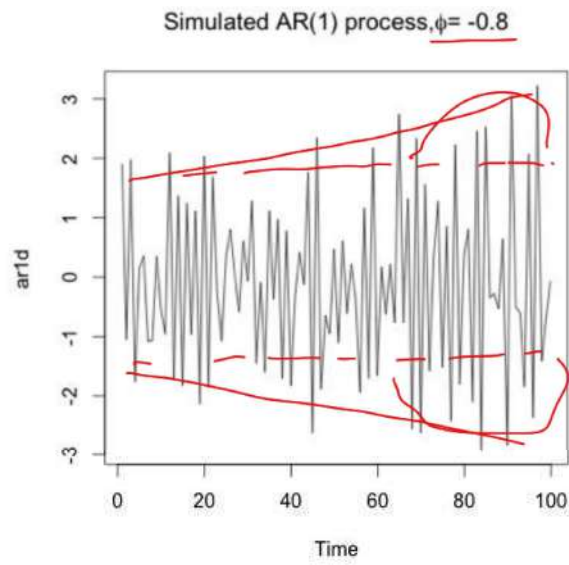
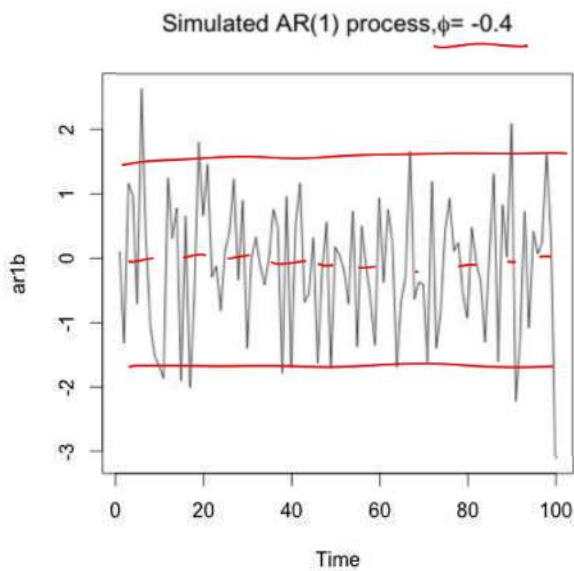


Note that all of the simulated AR(1) processes, barring the last one here, would be classified as stationary. However, as the value of the coefficient ϕ_1 approaches 1, the resulting time series becomes increasingly non-stationary in mean (i.e. an increasing trend becomes evident). For $\phi_1 = 1$ you get a process referred to as a “random walk”.

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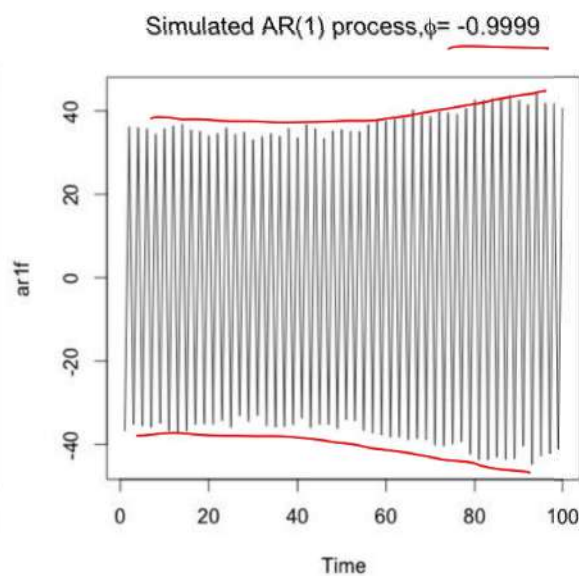
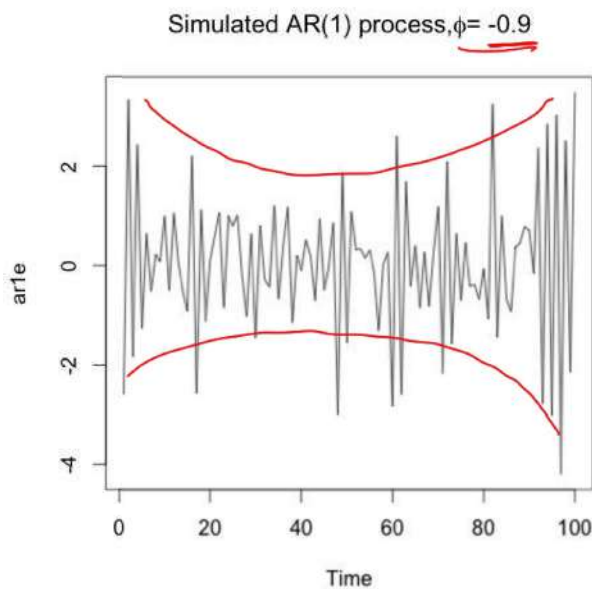
Autoregressive, AR(p), models

ϕ , negative



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Autoregressive, AR(p), models



As the value of ϕ_1 approaches -1, the resulting time series becomes more non-stationary in its variance. For $\phi_1 = -1$, you get a process called a “random jump” which has a changing variance and is hence non-stationary. For $|\phi_1| > 1$ the mean increases exponentially and hence the AR process is non-stationary.

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Autoregressive, AR(p), models

$$E(X_t) = E(c + \phi_1 X_{t-1} + \epsilon_t)$$

$$E(\phi_1 X_{t-1}) = \phi_1 E(X_{t-1})$$

$$\epsilon_t \sim N(0, \sigma^2)$$

Properties of a stationary AR(1) process:

1. The mean of a stationary AR(1) process is given by:

$$E[X_t] = E(c + \phi_1 X_{t-1} + \epsilon_t) = c + \phi_1 E[X_{t-1}] + 0$$

Since the process is stationary, we have that $E[X_t] = E[X_{t-1}] = \mu$

$$\Rightarrow E[X_t] = \mu = c + \phi_1 \mu$$

Therefore,

$$\mu - \phi_1 \mu = c \rightarrow \mu(1 - \phi_1) = c$$

and finally:

$$\mu = \frac{c}{1 - \phi_1}$$

For the case where $c = 0$, the above simply yields $E[X_t] = 0$. The corollary of this is also true – for AR(p) processes where the mean = 0, $c = 0$.

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Autoregressive, AR(p), models

Properties of a stationary AR(1) process:

$$V(\phi_1 X_{t-1}) = \phi_1^2 V(X_{t-1})$$

2. The variance of a stationary AR(1) process:

$$V[X_t] = V(c + \phi_1 X_{t-1} + \epsilon_t) = 0 + \phi_1^2 V[X_{t-1}] + \sigma^2$$

Since the process is stationary, we have that $V(X_t) = V(X_{t-1})$, and hence

$$V(X_t) - \phi_1^2 V(X_t) = \sigma^2$$

and finally:

$$V[X_t] = \frac{\sigma^2}{1 - \phi_1^2}$$

3. The ACF of a stationary AR(1) process: For the sake of simplicity, we consider an AR(1) process with mean 0 (i.e. $c = 0$):

$$X_t = \phi_1 X_{t-1} + \epsilon_t$$

If X_t is stationary, then there is only the autocorrelation at lag $g=1$: $\rho(1) = \phi_1$

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Autoregressive, AR(p), models

$$\rho(h) = \phi_1^h$$

$$\rho(3) = \phi_1^3 < \phi_1$$

acf

NB: Thus, the ACF of an AR(1) model 'decays' exponentially to 0 as the lag $g \rightarrow \infty$

The following slides show some examples of the ACF of a stationary AR(1) process



Properties of a **stationary AR(p) process**:

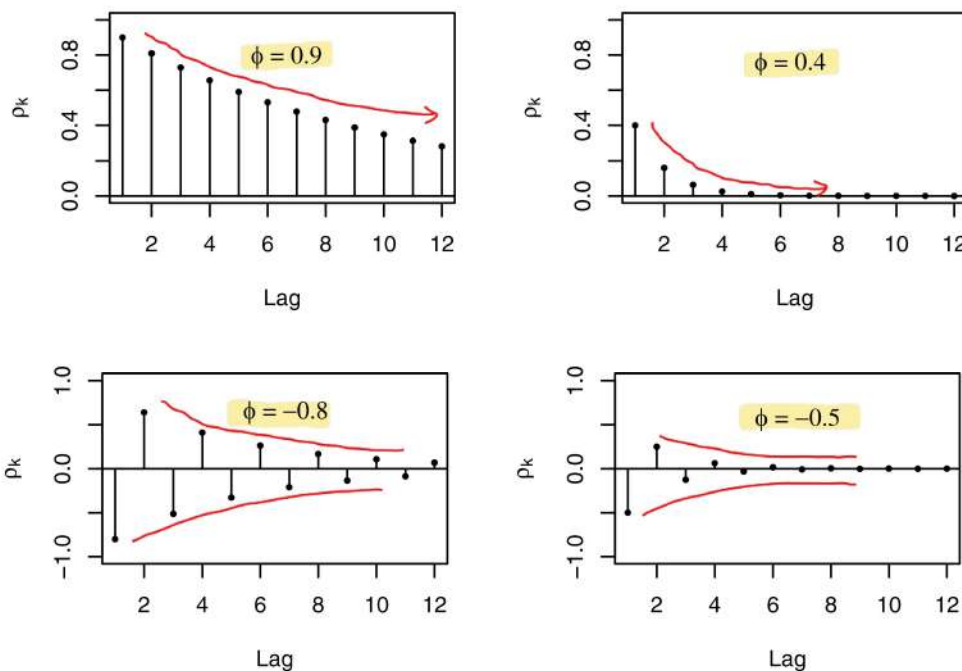
We can extend some of these results to an AR(p) process, where we obtain:

$$E[X_t] = \mu = \frac{c}{1 - \phi_1 - \phi_2 - \dots - \phi_p}$$

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Autoregressive, AR(p), models

Exhibit 4.12 Autocorrelation Functions for Several AR(1) Models



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Autoregressive, AR(p), models

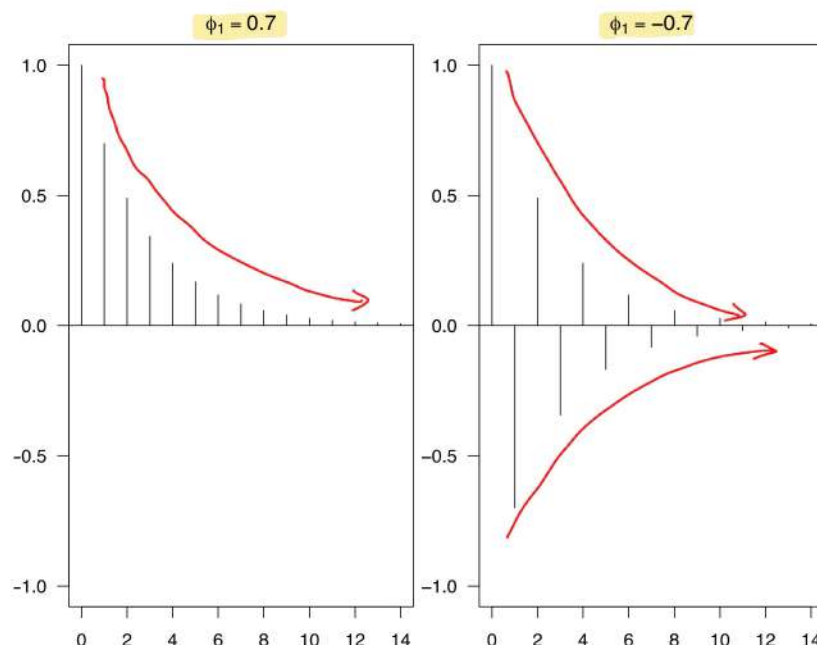


Figure 3.1: Autocorrelation functions for two of AR(1) models.

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Moving Average, MA(q), models

Rather than using past values of the forecast variable in a regression, a moving average model uses past forecast errors in a regression-like model. A process $\{X_t\}$ is said to be a moving average process of order q (abbreviated MA(q)) if:

$$X_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$

Where ϵ_t represents a white noise process (i.e. a series of independently and identically distributed values with mean 0 and variance σ^2)

In a MA(q) process, the current value of the series is a weighted average of past "white noise" terms. Notice that each value of X_t can be thought of as a weighted moving average of the past few 'forecast' errors.

- **NB NOTE:** The moving average MA(q) process defined above is **NOT** the same thing as the moving average smoothing technique! *Difference?*
 - The moving average smoothing technique, MA(k), is applied to a time series to smooth out random variation (or a seasonal component and random variation in the context of a Classical decomposition).
 - Here, the moving average process, MA(q) is seeking to model the dependency structure in a time series

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Moving Average, MA(q), models

A moving average model is usually used to model a time series that shows short-term dependencies between successive observations. They are important because:

1. They are very useful in evaluating the effect of random events (e.g. natural disasters) on a wide variety of economic processes.
2. Any q-correlated process (i.e. any process correlated with its past q values) can be represented as a moving average process.
3. They are widely used in conjunction with autoregressive processes to model a vast range of time series processes.

So, what's the big difference between MA(q) and AR(p) processes?

- An AR(p) model includes lagged terms of the time series itself as explanatory variables, while a MA(q) model includes lagged terms on the forecast errors/random component.
 $X_{t-i}, i=1, \dots, p$
 $\epsilon_{t-j}, j=1, \dots, q$
- In an autoregressive process, the actual values of the time series are used to form the next realization. In a moving average process, only the random component/forecast errors are used.
- Autoregressive processes depend upon the entire set of observations before time t. The moving average process depends directly on the forecast errors, and so the observations more than q periods in the past provide no information to the MA(q) process and can be ignored.

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Moving Average, MA(q), models

Properties of MA(q) processes

1. Mean:

$$\rightarrow E[X_t] = E(c) + E[\epsilon_t] + \theta_1 E[\epsilon_{t-1}] + \theta_2 E[\epsilon_{t-2}] + \dots + \theta_q E[\epsilon_{t-q}] = c$$

as all the random errors have mean 0.

$$= \sigma^2 + \theta_1^2 \sigma^2 + \theta_2^2 \sigma^2 + \dots + \theta_q^2 \sigma^2$$

2. Variance:

$$V[X_t] = \sigma^2(1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2)$$

3. Stationarity: MA(q) models are always stationary because they are finite linear combinations of a white noise sequence for which the mean and variance are constant (i.e. they do not depend on time)

MA(2): acf

4. A MA(q) series is only linearly related to its first q lagged values and hence is a "finite-memory" model.

$$X_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$

5. The ACF of a MA(q) process has the distinctive feature of vanishing at lags greater than q:

This is because the observed value in time period t, X_t , is related to only the previous q random errors. Since the autocorrelation is a measure of linear association (dependence) it makes sense that it would be equal to 0 at lags greater than q.

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Moving Average, MA(q), models

Properties of the MA(1) process

The simplest example of an MA process is the first-order case, denoted MA(1), given by:

$$X_t = c + \epsilon_t + \theta_1 \epsilon_{t-1}$$

1. The mean of a MA(1) process: $E[X_t] = E[c + \epsilon_t + \theta_1 \epsilon_{t-1}] = c + 0 + \theta_1 \times 0 = c$

2. The variance of a MA(1) process: $V[X_t] = \sigma^2(1 + \theta_1^2)$

The MA(1) process has an autocorrelation of 0 more than 1 lag away, as the process has a memory of only 1 period and “forgets” what happened in the further past. In theory, this implies that future forecasts only need very recent information to make predictions.

3. The autocorrelation function is given by:

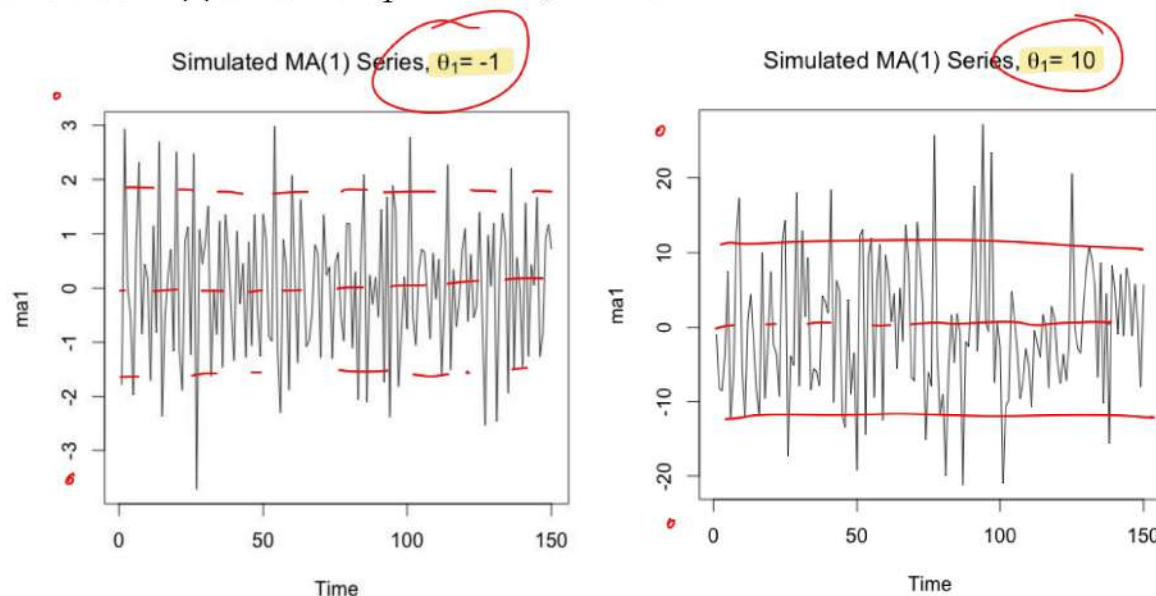
$$\rho(1) = \frac{\theta_1}{1 + \theta_1^2}$$

The autocorrelation plot thus shows a single non-zero autocorrelation at lag 1, with 0 autocorrelations at all higher lags.

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Model choice: Using the ACF/PACF

Simulated MA(1) series with $\theta_1 = -1$ & 10, $n = 150$:



Note that only the scale of the values change as the magnitude of the coefficient θ_1 changes. Both are plausibly stationary time series.

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Entire ARIMA modelling process overview:

When fitting an ARIMA model to a set of time series data, the following procedure provides a useful general approach:

- 1) Plot the data and identify any unusual observations.
- 2) If necessary, transform the data (using a Box-Cox transformation) to stabilize the variance.
- 3) If the data are non-stationary, take first differences of the data until the data are stationary.
- 4) Examine the ACF/PACF: Is an ARIMA(p,d,0) or ARIMA(0,d,q) model appropriate?
- 5) Try your chosen model(s) and use the AIC to search for a better model.
- 6) Check the residuals from your chosen model by plotting the ACF of the residuals and doing the diagnostic tests of the residuals. If they do not look like white noise, try a modified model.
- 7) Once the residuals look like white noise, calculate forecasts.